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RESEARCH NOTE

Palaeomagnetism of Proterozoic dykes from Bornholm, Denmark

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SUMMARY

The island of Bornholm (Denmark) is a structural horst situated at a kink on the margin of and linking two major fault zones, the Sorgenfrei–Tornquist Zone (STZ) and the Tornquist–Teisseyre Zone (TTZ), the fault zones of which separate the downfaulted Danish–Polish Basin from the Baltic Shield.

Thirteen dolerite dykes, out of a swarm of more than 200, all belonging to the supposed Precambrian N–S to NE–SW striking dyke swarm on Bornholm, have been sampled and palaeomagnetically investigated. The dykes vary in thickness between 0.2 and 60 m and are intruded in Precambrian gneisses and granites of Gothian and/or Subjotnian (1650–1280 Ma) age. Principal component analysis of detailed AF and some thermal demagnetization experiments reveal three groups of stable direction clusterings from the dykes investigated.

The stable remanent magnetization of the major dyke Keldseå dyke and the baked gneiss contact corresponds to a palaeomagnetic mean pole (group A). Comparing it with the APW curve for Fennoscandia, an initial cooling age for this dyke of around 1360 Ma (Subjotnian age) is suggested. The result is confirmed by dual polarities and a positive baked contact test. Nine other dykes show a magnetization corresponding to a younger group of poles (group B), which cluster on the APW path around 1200 Ma (Jotnian age). This result is also confirmed by a positive contact test. Two poles (group C) fall on the APW path around 950 Ma (Sveconorwegian age). Remaining directions display a distribution of poles on the sphere, which suggest the presence of either late Proterozoic–Cambrian overprints or injections of some younger dykes. Based upon a total of 16 Bornholm dykes hitherto studied, the polarity frequency ratio N/R is 7/9.

The remanent magnetizations and equivalent palaeomagnetic poles found in the Bornholm dyke swarm thus appear to indicate at least three different periods of igneous activity with dyke injections in early Proterozoic time for this otherwise homogeneously looking group of dykes from the margin of the Baltic Shield.

Key words: APWP, Baltica, Bornholm, dykes, palaeomagnetism, Proterozoic.

INTRODUCTION

The 600 km^2 Danish island of Bornholm (latitude 55° N, longitude 15° E), is situated in the southern part of the Baltic Sea, between Sweden and Poland (Fig. 1). Bornholm is a rhomb-shaped basement horst bounded by N–S and WNW–ESE striking faults. Its geology is closely related to that of southernmost Sweden.

Bornholm belongs to the southern margin of the Fennoscandian Shield, and is situated like a cornerstone just ENE of the offset junction, separating the two prominent WNW-trending fault zones, the STZ (Sorgenfrei–Tornquist Zone) and the Tornquist–Teisseyre Zone (TTZ), which mark the transition between the downfaulted Danish–Polish sedimentary basin and the Baltic Shield.

Previously published palaeomagnetic results from Bornholm are limited to data from four dolerite dykes, including the major dyke at Listed (Abrahamsen 1977), and to data

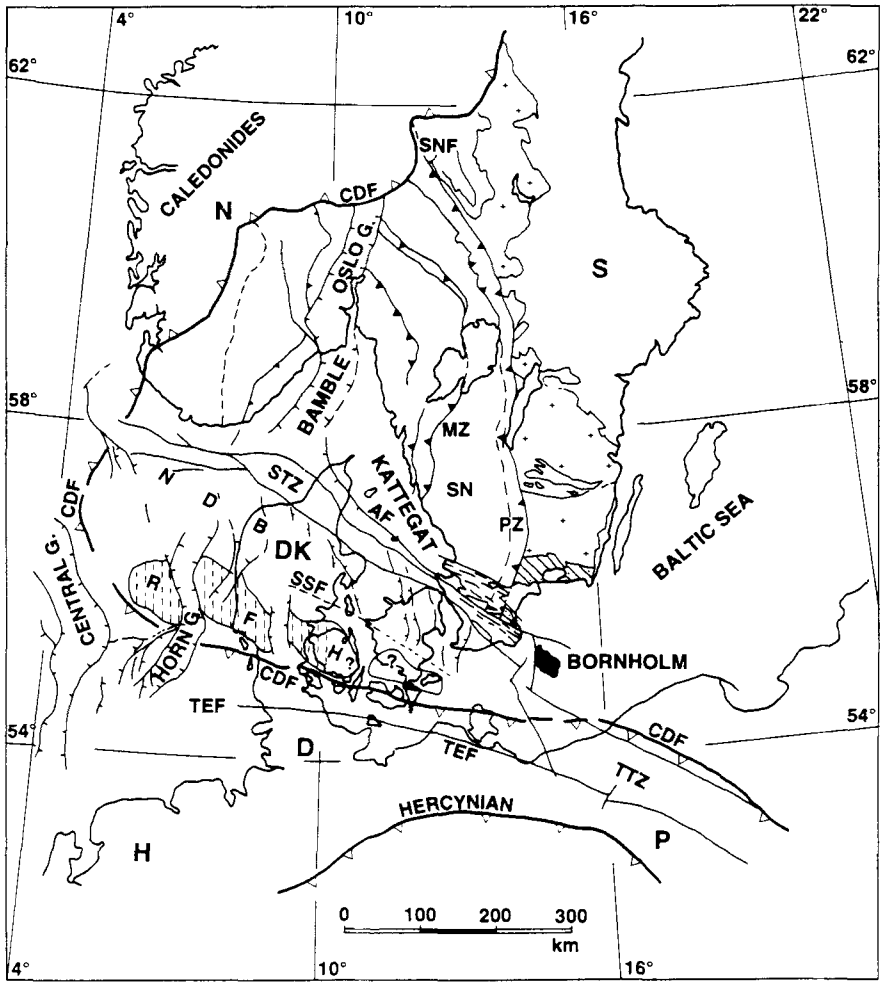


Figure 1. Index map of the Danish–Polish, south Scandinavia–Baltic area. AH, Anholt-Frederikshavn Fault; CDF, Caledonian Deformation Front; MZ, Mylonite Zone; NDB, Norwegian Danish Basin; PZ, Protogine Zone; RFH, Ringkøbing Fyn High; SNF, Sveconorwegian Front; SN, Sveconorwegian orogen; SSF, Silkeborg Samsø Fault; TEF, Trans European Fault (modified from the EUGENO-S Working Group 1988).

from the Lower Cambrian Nexø Sandstone (Prasad & Sharma 1978, 1980). Recently, more than 500 individually oriented drill cores and hand samples from 13 dolerite dykes (14 sites), from the Precambrian granites and gneisses, and from the Lower Cambrian to Ordovician sediments (sandstones, shales and limestones) were collected for palaeomagnetic studies and palaeogeographical reconstruction purposes.

The purpose of the present contribution is to find out whether or not Bornholm acted as an independent block with respect to Baltica in early Proterozoic time, based upon some new palaeomagnetic results. All sites investigated are located on the well-exposed north-east coast, as shown in Fig. 2. Detailed results from the early Palaeozoic sediments will be presented elsewhere (Lewandowski, M. & Abrahamsen, N., in preparation).

GEOLOGY

The fault zones STZ and TTZ, which have a long and complicated tectonic history (Tornquist 1908; Sorgenfrei 1969; Pegrum 1984; Liboriussen, Ashton & Tygesen 1987; Ziegler 1982, 1988), separate the uplifted Fennoscandian Shield from the Danish–Polish sedimentary basin, which contains a sequence of up to 10 km or more of Phanerozoic sediments (e.g. Pozaryski & Brochwicz-Lewinski 1978; Brochwicz-Lewinski, Pozaryski & Tomczyk 1981; Guterch *et al.* 1986; Pegrum 1984; Madirazza *et al.* 1990). A recent series of overviews and guides of the geology of Bornholm was published by Surlyk (1980) and Berthelsen (1980, 1989, 1992a,b, 1993).

About three-quarters of the Pre-Quaternary surface of Bornholm consists of Precambrian (Proterozoic) gneisses and granite (Noe-Nygaard 1963), while the rest is covered by Palaeozoic and Mesozoic sediments. The bedrock is poorly exposed because of glacial drift, but along the north-east coast exposures are excellent.

An early geosynclinal stage, with deposition of sediments and perhaps volcanics (Michelsen 1961; Berthelsen 1989), was followed by metamorphism and development of the Bornholm Gneisses and the ?older Rønne Granite of Gothian age (1780–1650 Ma, Fig. 2), the basement evolution being closely related to that of the south Swedish Blekinge basement (e.g. Johansson & Larsen 1989). Younger granites (Vang, Hammer, Almindingen and Svaneke) were formed around 1400 Ma (Fig. 2), and afterwards elevated and cooled down to some 300 °C between 1350 and 1250 Ma, as evidenced by K–Ar ages on biotites. Further slow cooling

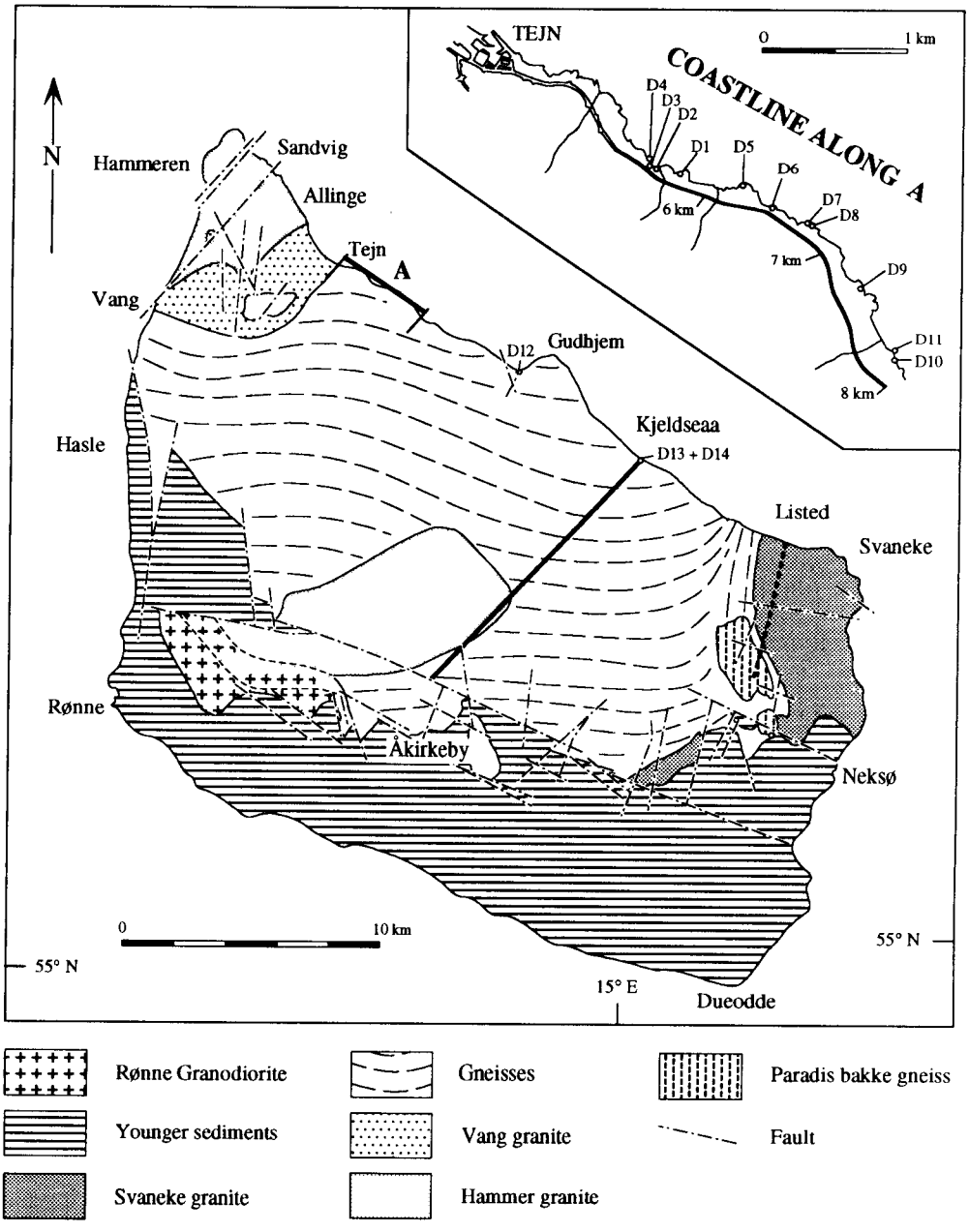


Figure 2. Geological map of Bornholm (modified from Abrahamsen 1977). The palaeomagnetic sampling sites are all situated along the north-east coast, as indicated with dyke site numbers D1 through D14 (enlarged section at top right).

from some 200 to 100 °C between 950 and 800 Ma are evidenced by the Ar–Ar spectra (Holm 1985). An eastward translation of Bornholm relative to central south Sweden (Falkum 1985; Berthelsen 1989; Park, Åhäll & Boland 1991) may have taken place before the Sveconorvegian deformation (1100–950 Ma), leaving the island unaffected by this orogenic phase (Fig. 1). After formation of the crystalline basement, a cratogenic period with faulting and injection of many dolerite dykes followed (in one or more generations). All gneisses and granites are Precambrian, being overlain by Lower Cambrian Nexø Sandstone and younger sediments.

Clastic sediments and carbonates were deposited in Palaeozoic time and are now preserved in a mosaic of downfaulted blocks on southern Bornholm (Fig. 2). The former thickness of the sediments along the north-east coast is not known, but on southern Bornholm stable oxygen isotopes show that deep burial diagenesis took place in the Cambro–Ordovician limestones with closing temperatures in the range between 90 and 150 °C, corresponding to burial depths of 2–3 km (Buchardt & Nielsen 1984).

In Bornholm no dykes have been observed cutting the (?)Lower Cambrian Nexø Sandstone or younger sediments, but in nearby south Sweden post-Silurian NE striking dykes, Permian NW and WNW striking dykes, as well as Tertiary volcanic plugs, are known (Bylund 1974; Klingspor 1976; Bylund & Pesonen 1987). At other places along the STZ, Phanerozoic volcanism is also known (e.g. Thorning & Abrahamsen 1980; DGU 1981; Abrahamsen *et al.* 1992).

Precambrian dykes are numerous in Sweden, and palaeomagnetic results were recently compiled (Pesonen *et al.* 1989, 1991; Torsvik *et al.* 1992). Furthermore, Bylund (1992) reported upon substantial remagnetization of some dykes in the Protogene Zone (South Sweden) during the Sveconorwegian orogeny.

The Bornholm swarm includes at least 254 dykes (Münther 1945, 1973). About 200 dykes have a combined width of 400 m over an exposed length of 30 km along the north-east coast, which is about 1.3 per cent of the basement volume at the now exposed level. Only four dykes more than 10 m wide are known in Bornholm: the Keldseå dyke (60 m wide), the Kås dyke (40 m), the Listed dyke (31 m) and the Salne dyke (22 m). These four dykes alone account for about 38 per cent of the crustal expansion by the dyke swarm.

The thickest dolerite dyke, the Keldseå Dyke, has given a K–Ar whole rock age of 1000–1100 Ma (Larsen 1980). The Listed dyke and the surrounding Svaneke granite are cut by several clastic 'dykes' of sandstone up to 1.3 m wide (Ussing 1899; Bruun-Petersen 1975). The sediment of these clastic dykes is correlated with the (?)Lower Cambrian Nexø Sandstone and therefore date at least the Listed dolerite dyke as pre-Lower Cambrian. The clastic dykes furthermore constrain the present-day exposed erosional level of the Listed dyke and granite to be situated within a few tens of metres of the Eocambrian erosional peneplain of the area.

The dolerite dykes in Bornholm are typically close to vertical and the majority strike between N and NE. Other directions are also found, which may be younger. One of these dykes show a Permian magnetization (Abrahamsen & Lewandowski 1992).

The fault pattern of Bornholm is quite complicated, and at least four periods of horizontal movements along vertical faults have been noted (Münther 1945). In decreasing relative age they took place along faults striking (1) N, (2) NE, (3) N or NNE and (4) NE. The sequential fault pattern and hence shifting stress directions with time may suggest that dyke injections could have taken place at more than one time.

PALAEOMAGNETIC INVESTIGATIONS

In situ magnetic susceptibility was measured by a portable susceptibility meter (Geofyzika Brno: Kappameter KT-5) on smooth, exposed surfaces of the dykes as well as on the surrounding gneiss, the results of which are summarized in Table 1. The maximum range in susceptibility was found among the four major dykes, the lowest value being in the 22 m wide and strongly altered dyke at Salne Bay (site 12), whereas the highest value is seen in the central part of the 60 m wide Keldseå dyke (site 13). The overall mean susceptibility of the dykes (24×10^{-3} SI) is only slightly higher than that of the gneiss (20×10^{-3} SI).

The NRM logarithmic mean intensity is also compiled in Table 1. A variation of more than two orders of magnitude is found within the individual dykes 5, 9 and 11, and more than one order of magnitude within dykes 4, 12 and 13/14. Thus, although a dyke may look homogeneous to the naked eye, the NRM intensity varies drastically within individual dykes. The modified Q ratio (NRM/susceptibility) varies between 0.4 and 3.7, except for the central part of the Keldseå dyke (site 13), which has an extraordinarily high Q ratio of 30.

Table 1. Susceptibility and NRM intensity of dykes and gneiss.

SITE	WIDTH m	k No. of Sus meas	k Mean-Sus 10-3 SI	k sd 10-3 SI	J No. of NRM meas.	J NRM mA/m (Log)	Q J*4pi/k
DYKE1	40	12	28.5	4.6	13	928	2.6
DYKE2	0.5				3	278	
DYKE3	3	30	5.6	4.5	2	48	0.7
DYKE4	1	27	14.3	9.3	2	263	1.5
DYKE5	0.5	12	22.2	7.4	7	108	0.4
DYKE6	0.5	18	24.4	5.4	8	1127	3.7
DYKE7	>3.5	18	34.0	5.9	7	874	2.0
DYKE8	1	21	26.2	6.3	10	1054	3.2
DYKE9	>3.5	20	33.6	9.3	17	1112	2.6
DYKE10	0.3	11	21.7	11.7			
DYKE11	2x0.5	21	27.0	6.9	11	377	1.1
DYKE12	22	37	0.4	0.2	5	4	0.9
DYKE13c	60	13	42.1	6.9	6	*15877	30.0
DYKE14	60	12	31.5	4.6	19	758	1.9
DYKE15	31						
MEAN All Dykes		13(252)	24.0	11.1			
GnD1		55	14.6	6.1			
GnD3		20	15.8	7.5			
GnD4		36	20.5	5.5			
GnD5		38	17.8	4.5			
GnD6		40	26.4	3.9			
GnD7		40	15,5	5.6			
GnD8		39	5.1	2.7			
GnD9		129	25.0	5.8	7	282	0.9
GnD10		40	23.1	5.0			
GnD11		39	27.8	3.9	2	200	0.6
GnD12		40	21.0	9.6			
GnD13/14		39	27.9	3.0	8	472	1.3
GnD15							
MEAN All Gneiss		12(555)	20.0	6.4			

A total of 173 individually oriented hand samples and/or drilled cores were collected from 13 dolerite dykes (sites 1–14) of the NE–SW striking Bornholm dykes of supposed Precambrian age, as well as from three baked contacts of the host rock gneiss. At least eight independently oriented samples or cores were usually obtained at each dyke site, although not all were later processed successfully in the laboratory (Table 2). Field cores were oriented by means of a sun compass, whereas the hand samples were mostly oriented by a magnetic compass. Local magnetic variation was controlled by sun compass or topographical bearings.

Magnetic remanence measurements were made on a Molspin spinner magnetometer. A total of 102 specimens were stepwise demagnetized in AF fields in steps of 5 mT up to 40 mT, and further in 10 mT steps up to 100 mT. After the AF treatment a few samples were demagnetized thermally.

The NRM directions of all specimens measured ($N = 102$) are shown in stereographic projection in Fig. 3(a). A cluster around the present-day geomagnetic field direction ($D_0 = 0°$, $I_0 = +70°$) is evident, indicating the presence of a strong viscous overprint. The distribution is elongated, however, suggesting that non-viscous components are also preserved,

Table 2. Bornholm palaeomagnetic results.

SITE NAME, Width	Type of CHRM	No of samples N:N1:N2:N3	DEC o	INC o	a95 o	k	R	Lat oN	Long oE	dp o	dm o	p o	Polarity N/M/R	Pole in Fig. 7	Proposed Pm age (Ma) Reference
DYKE1, KÅS, 40m	B	15:13:10:9	87	73	15	12	8.3333	46	62	25	27	59	R	1	900-950
DYKE2, 0.5m	B+A	8:3:4:4	9	-20	17	30	3.9000	-24	6	9	18	100	R	2	(?1280)
DYKE3, 3m	A	8:2:3:3	8	49	17	54	2.9630	64	178	15	22	30	N	3	?(600-700)
DYKE4, 1m	B+A+V	8:2:3:3	218	79	27	21	2.9048	37	359	50	52	69	R	4	?1100-1200
DYKE5, 0.5m	B	8:6:4:3	36	-21	15	69	2.9710	-18	338	8	16	101	R	5	ca. 1200
DYKE6, 0.5m	B	8:8:9:7	41	-45	4.9	151	6.9603	-1	339	4	6	117	R	6	ca. 1200
DYKE7, >3.5m	B	8:7:6:6	167	35	7.4	82	5.9390	-15	27	5	9	19	R	7	?remag (600-800?)
	B,e,th	8:7:5:5	256	-76	16	22	4.8182	53	59	28	30	154	R	7Th	900-950
DYKE8, 1m	B,e	10:10:10:10	165	-25	18.7	7.5	8.0001	47	217	11	20	103	N	8	?(600-700)
	B,e,th	10:10:6:6	182	-42	14	24	5.7917	59	191	11	17	114	N	8Th	?(600-700)
DYKE9, >3.5m	B	11:10:20:17	33	-45	5.4	44	16.6364	-4	346	4	7	116	R	9	ca. 1200
Gneiss baked by Dyke9	B+A	6:2:3:3	34	-47	23	29.4	2.9309	-2	346	19	30	118	R	Gn9b	+contact test
DYKE10, 0.3m	Not treated	5:0:0:0													
DYKE11, 2x0.5m	B,sel	8:3:7:5	250	63	18	20	4.8001	25	327	22	28	44	R	11	ca. 1200
DYKE12, SALNE, 22m	B+A	13:4:2:2	49	-2	-	-	-	-21	321	-	-	91	R	12	ca. 1200
KELDSEÅ,60m															
Site 13 (centre)	B	6:6:6:6	23	46	5.3	159	5.9686	58	154	4	7	27	M	13	ca. 1360
Site 14 (margin)	B,e	7:7:19:19	31	40	3.9	74	18.7568	50	147	3	5	23	N	14	ca. 1360
Gneiss baked by 13/14	B+A	3:3:6:6	30	41	13	27.3	5.8148	51	148	10	16	23	N	Gn14b	+contact test
LISTED, 31m	B	44:44:32:32	307	-14	4.1	39	31.2050	14	250	2	4	97	N	Li	Ref. (1); ?(700-800)
VIGEHAVN,0.3m	B	14:14:7:7	36	-26	6.1	97	6.9384	15	159	4	7	104	N	Vi	Ref. (1); ca. 1200
VASEAA, 1.5m	B	22:22:11:11	81	-44	8.9	27.6	10.6373	-16	128	6	10	116	N	Va	Ref. (1); ?1200
BØLSHAVN, 3m	B	35:35:9:9	42	-26	8.9	34	8.7657	13	153	5	10	103	N	Bø(A)	Ref. (1); ?1200
	A	35:35:6:6	333	-20	13.5	25.5	5.8045	21	224	7	14	100	N	Bø(B)	Ref. (1); 700-800 remag?
NEXØ Sandstone Fm	B	77:77:14:14	226	-30	11.5	13		38	134	7	11	106	R	NS1	Ref. (2); (?)early Cambrian
NEXØ Sandstone Fm								55	203				M	NS	Ref. (3); early Cambrian

Explanatory notes to Table 2:
N: Number of collected cores/samples, individually oriented in the field.
N1: Number of independently oriented specimens measured.
N2: Number of CHRM components obtained by the PCA.
N3: Number of CHRM results used in mean calculation and Fisher statistics.
sel: Selected data; for selection criteria see text.
th: Thermal demagnetization, otherwise A.F.
DEC=Declination; INC=Inclination; a95=Circle of 95% confidence.
V: Viscous or soft magnetic component (coercivity between 0 and 15 mT).
A: Intermediate component (coercivity between 15 and 50 mT).
B: Hard component (coercivity above 50 mT).
e: Endpoints at 100 mT A.F.; sel: Selected data.
k=Fisher precision parameter; R=Unit vector;
Lat=Apparent pole latitude; Long=Apparent pole longitude.
dp & dm=minor and major semiaxis of 95% ovals of confidence.
p=Apparent palaeomagnetic pole co-latitude.
Ref. (1): Abrahamsen 1977. (2): Prasad & Sharma 1980. (3): Lewandowski & Abrahamsen (in prep. for Geophys.J.Internat.).

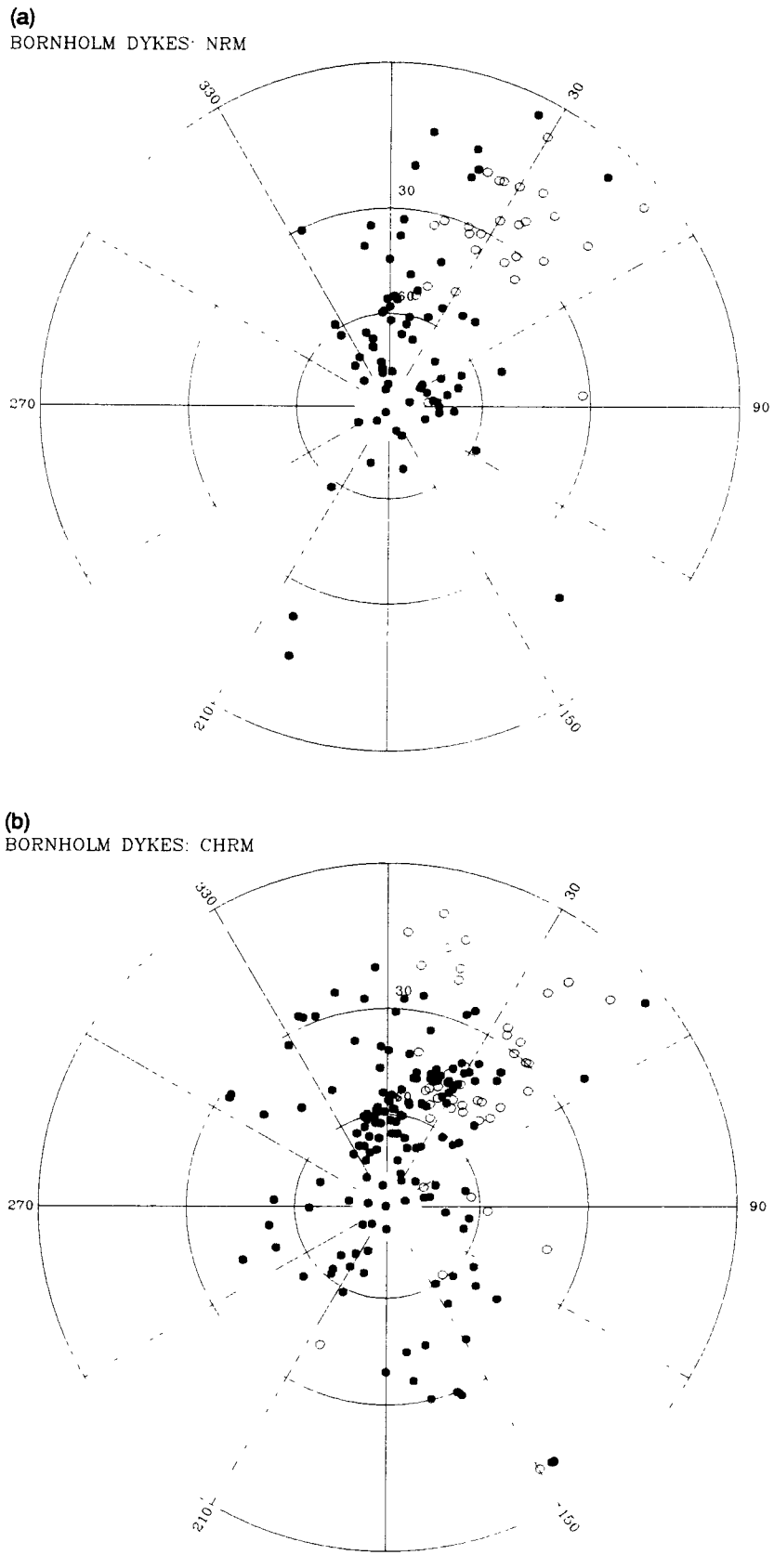


Figure 3. (a) NRM of all specimens measured ($N = 102$). Two clusters are seen, one with positive inclinations (solid dots) around the present-day field direction ($D = 0°$, $I = +70°$), and one with negative (open circles) and more shallow inclinations around ($D = 30°$, $I = -30°$) (stereographic projection). (b) AF cleaned ChRM (characteristic remanent magnetization) directions with angular standard deviations less than 5° as found by PCA (principal component analysis). Both polarities are present, with normal polarities dominating (stereographic projection).

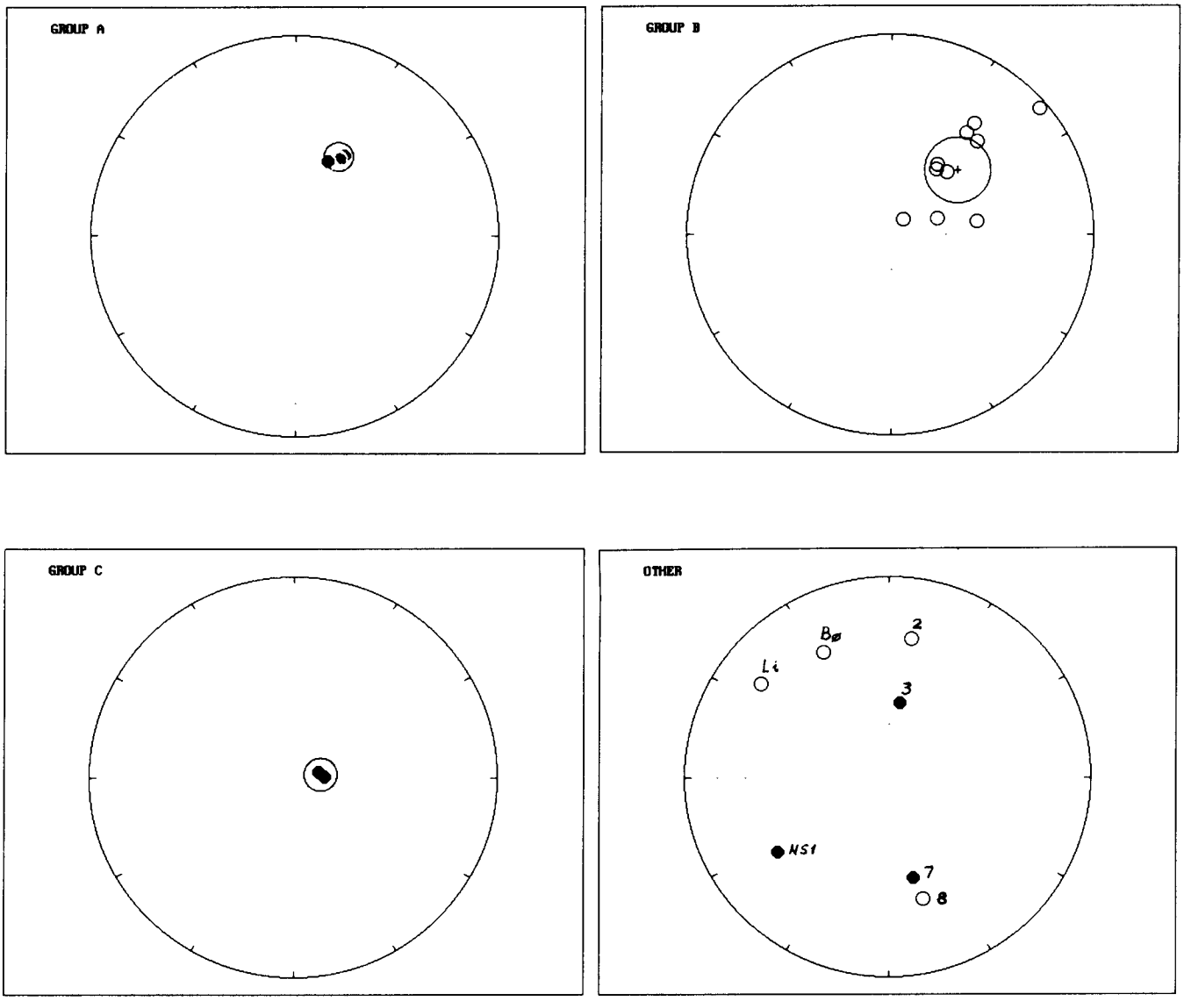


Figure 4. Cleaned mean directions with α_{95} confidence circles (*cf.* Table 2). Three magnetic directions are well defined in Group A (dyke 13/14 (Keldseå dyke) and its baked gneiss contact), Group B (dykes 4, 5, 6, 9, 11, 12, Vi, Va, Bø(A) and baked gneiss contact at dyke 9), and Group C (Kås and dyke 7), whereas the remaining dykes show different directions, possibly due to younger ages or due to remagnetizations (stereographic projection).

and indeed 27 of the specimens show negative inclinations, with a clustering around ($D = 30°$, $I = -30°$).

After AF cleaning, characteristic remanent magnetization (ChRM) components were defined by using least-squares principal component vector analysis (PCA), as described by Kirschvink (1980; see also Lewandowski, Jelenska & Halvorsen 1987). Choosing a value of 5° for the ASD (angular standard deviation) through at least four data points, ChRM components were obtained as shown in Fig. 3(b). After the AF cleaning there is still a cluster close to present-day directions, but other clusters with both positive and negative inclinations are now seen as well.

Cleaned mean directions are shown in the stereograms of Fig. 4, including the 95 per cent confidence circles (details are given in Table 2). It appears, that about two-thirds of the stable magnetic directions are well defined and may be sorted into three groups: Group A (dyke 13/14 (Keldseå dyke) and its baked gneiss contact), Group B (dykes 4, 5, 6, 9, 11, 12, Vi, Va, Bø(A) and baked gneiss contact at dyke 9), and Group C (the Kås dyke and dyke 7), whereas the remaining dykes show different directions, possibly due to either younger intrusion ages or remagnetizations (Li, Bø(B), dykes 2, 3, 7, 8). A result from the (?) Lower Cambrian Nexø Sandstone Formation, NS1, (Prasad & Sharma 1980) is also shown in this figure.

Representative AF demagnetization curves are shown in Fig. 5. The MDF (median destructive field) typically ranges between 10 and 20 mT, although in site 14 (the fine-grained margin of the Keldseå dyke) the range of MDE is from 10 to 40 mT. In dyke 6 an increase in intensity at 10 mT of up to 14 per cent is seen, caused by a reversal in polarity at this field. In the baked gneiss in contact with dyke 9 (3.5 m wide), the MDF are much higher in most specimens (35–45 mT) as compared to the unbaked gneiss site

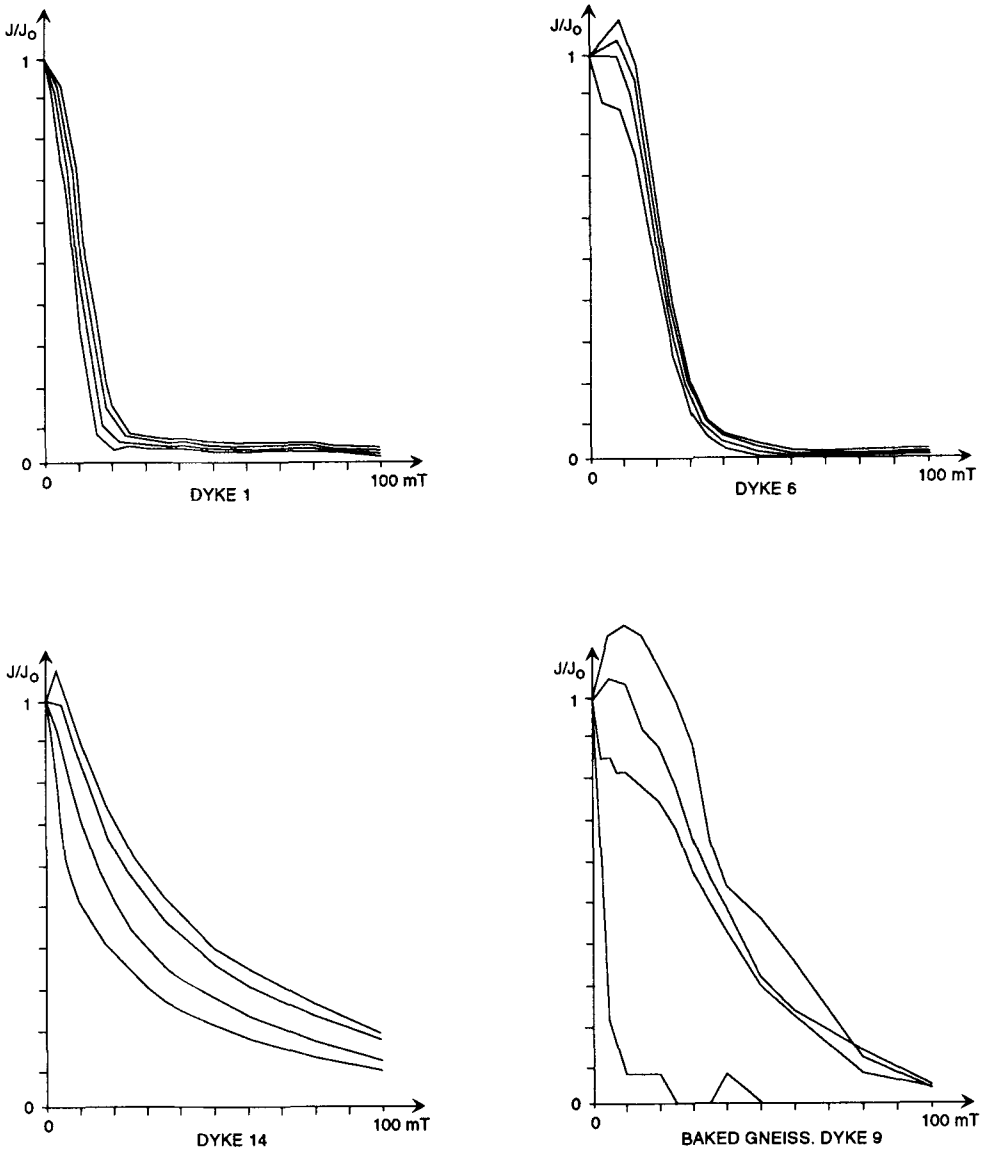


Figure 5. Examples (dykes 1, 7, 14 and the baked gneiss contact at dyke 9) of normalized intensity decay curves by AF demagnetization. The AF steps used in most cases were 0, 5, 10, 15, 20, 25, 30, 35, 40, 50, 60, 80 and 100 mT. The MDF (median destructive fields) are typically between 10 and 20 mT.

(*c.* 3 mT) 13 m from the contact. The low coercivities and the shape of the intensity decay suggest that the dominant remanence carriers are multidomain magnetites. Thermal demagnetization tests revealed blocking temperatures close to 580° (not shown), confirming that the dominating ferromagnetic minerals are magnetites.

In Fig. 6 typical examples of different directional changes during the AF demagnetizations are illustrated in stereograms and corresponding orthogonal projections. In most cases viscous components are removed between 0 and 10–15 mT, and the characteristic direction then remains stable and is well clustered up to 40 or 50 mT. Between 50 and 100 mT the directions are often more scattered, and some of the remaining remanence may be spurious in origin.

By the PCA analysis it was found that in most specimens three directionally distinct components with a characteristic range of coercivities could be isolated: a hard coercivity remanence with AF coercivities above 50 mT (typically between 50 and 100 mT, named the B component), an intermediate coercivity remanence with coercivities between 15 and 50 mT (named the A component), and a low coercivity or viscous component with coercivities between 0 and 15 mT (named the V component).

The results of the PCA analysis are compiled in Table 2, according to the three categories of coercivity, B, A and V. In some cases, a selection (indicated by 'sel' in Table 2) of the stable endpoints (at 100 mT AF demagnetization) was further undertaken if the remanence intensity was untypically low, if only three consecutive points were present in the line of the PCA, or if the orientation of a core drilled in the field was questionable.

PALAEOMAGNETIC RESULTS

The polarities of the Bornholm dykes also noted in Table 2, have been defined as 'normal' or 'reversed' in order to give

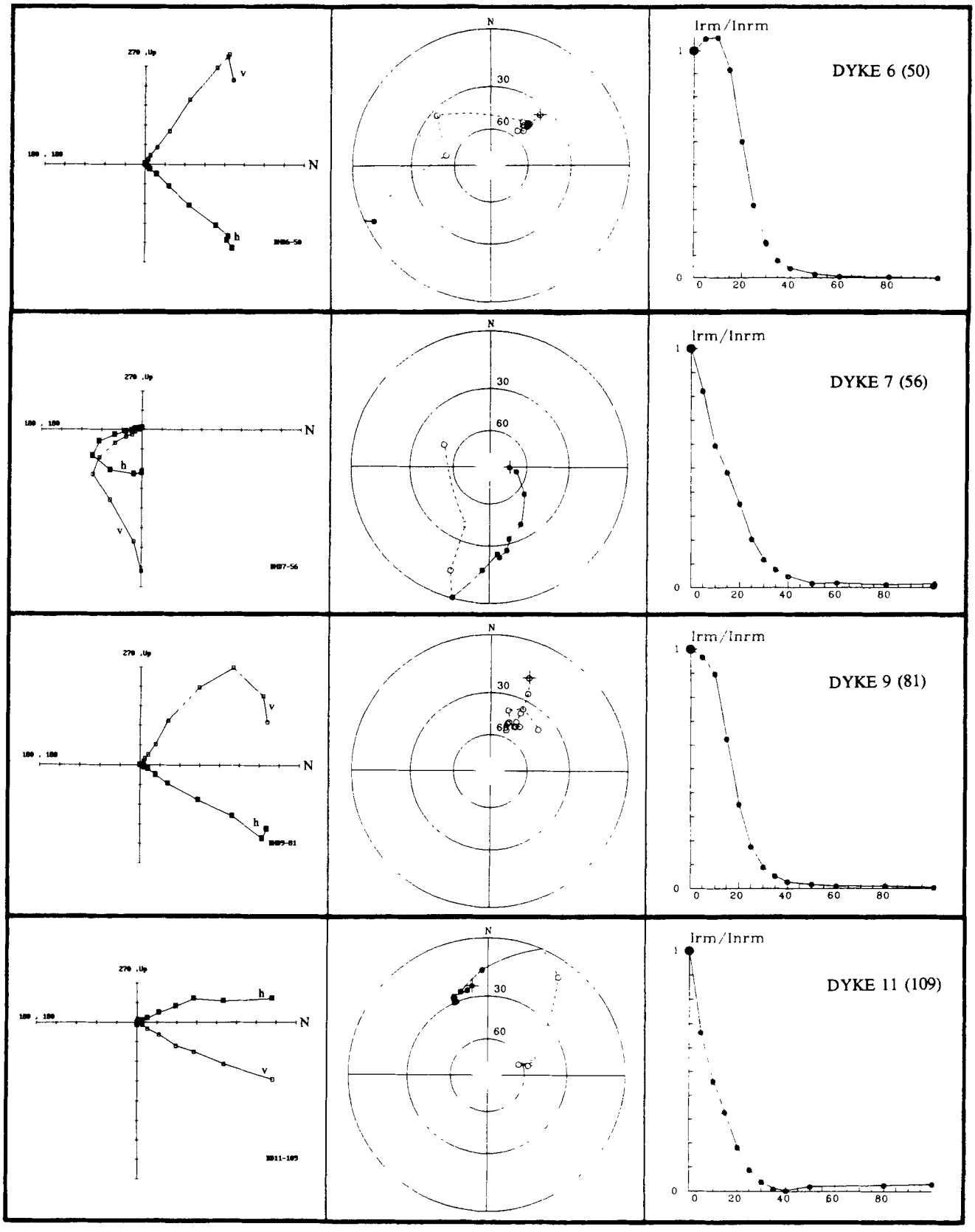


Figure 6. Stereograms (left), orthogonal plots (centre; scales in mA m $^{-1}$), and intensity decay curves (right; AF scale in mT) of individual specimens from dyke sites 1, 2, 5, 6, 7, 9, 11, 13 (both reverse and normal polarity), 14, and baked gneiss contacts at sites 9 and 14, showing characteristic changes during AF demagnetization.

(b)

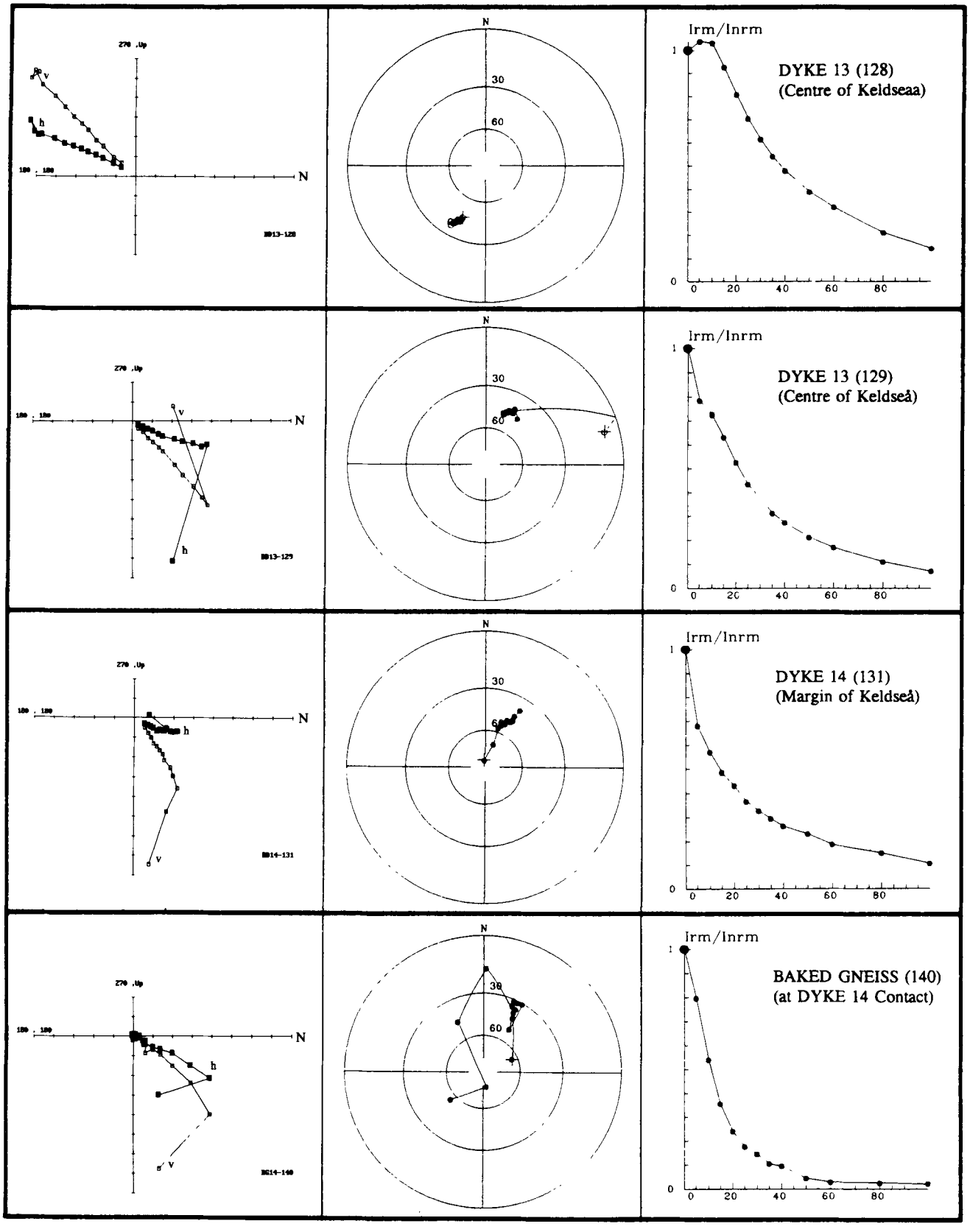


Figure 6. (*Continued.*)

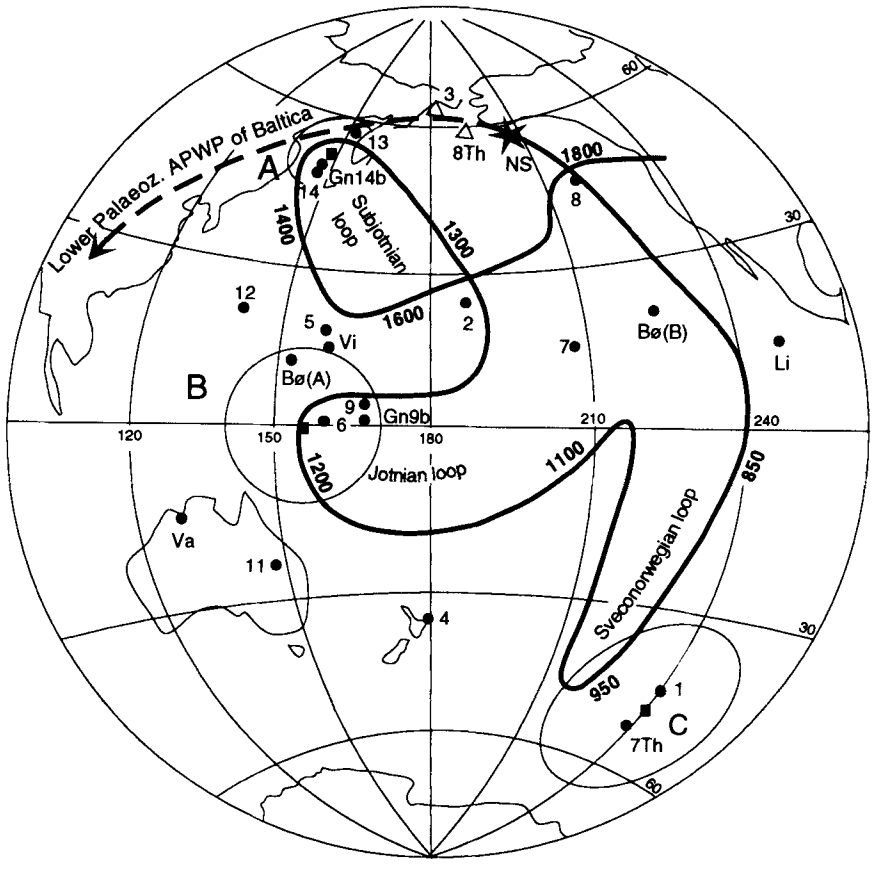


Figure 7. Palaeomagnetic results for Bornholm. The continuous curve is the Proterozoic APW path of Fennoscandia, calculated by means of the GMAP (Torsvik *et al.* 1990) from grand mean poles (Personen *et al.* 1989). Individual poles are from the Bornholm dykes and Lower Cambrian Nexø Sandstone (Table 2). Bornholm mean poles A, B and C (Table 3) are shown with 95 per cent errors (triangles are not included in the mean poles). Numbers on the APW curve indicate age in Ma.

the best fit with the Fennoscandian APW path (see Fig. 7). Of the 12 successfully treated dykes, nine are found to be of 'reverse' and three of 'normal' polarity (dykes 3, 8 and 13/14). Furthermore, at site 13 both polarities are found in the central part of the dyke (the slowly cooled 60 m wide Keldseå dyke). The previous study of four other dykes (Listed, Vaseå, Vigehavn and Bølshavn dykes) all showed 'normal' polarities (Abrahamsen 1977). Hence the polarity frequency ratio N/R is 7/9 (one being mixed), as based upon the 16 Bornholm dykes hitherto studied.

The apparent pole positions of the ChRMs of the Bornholm dykes (as labelled in Table 2) are plotted in Fig. 7, together with the Precambrian APW swath of Fennoscandia (plotted as north-seeking poles, as in Pesonen *et al.* 1989). The APW swath was constructed by means of the GMAP program by Torsvik, Smethurst & Pesonen (1990), using grand mean poles quoted in Pesonen *et al.* (1989). Also the palaeomagnetic poles for four other, previously studied, dolerite dykes from Bornholm (Abrahamsen 1977), and the Lower Cambrian Nexø Sandstone Formation (Lewandowski & Abrahamsen, in preparation) are shown.

The various apparent pole positions derived from the remanence mean directions (Fig. 4) of the individual Bornholm dykes studied fit remarkably well with the general APW path of Fennoscandia, and three clusters of poles from Bornholm, with mean poles, A, B and C may be calculated (Table 3), as shown in Fig. 7 with the corresponding 95 per cent circles of confidence. For the thin dykes a secular variation of the order of ±10° may be expected in the directions, but this effect is likely to be averaged out when calculating the mean poles:

Mean pole A is the mean of three poles: nos. 13 and 14 (the centre and margin of the big Keldseå dyke) and Gn14b (baked gneiss), which are well defined (A95 = 7.1°) and cluster closely west of Kamchatka on the peak of the Subjotnian Loop, suggesting a palaeomagnetic age of some 1360 Ma for this major dyke. The polarity is dominantly normal, but both polarities are found in the centre of the dyke (site 13), which may either be due to a prolonged cooling of the wide dyke, or it could be due to chemical alterations. A contact test (positive) of the surrounding baked gneiss confirms the primary nature of the normal polarity, the remanence of unbaked gneisss showing south-west downdip remanence and viscous overprints (Thomsen 1993). The quoted K–Ar age of 1000–1100 Ma (Larsen 1980) thus appears to be somewhat young, which may be due to Ar diffusion at moderately elevated temperatures during the Sveconorwegian orogeny. The positive baked contact test, combined with the dominant normal polarity, precludes the possibility of a late Carboniferous–early Permian overprint, as might otherwise be suggested by the proximity of European poles of these ages.

Table 3. Bornholm mean poles.

Site and Mean pole	Pole in Fig. 7	N	Dm	Im	R	a95	k	Lat oN	Long oE	A95 o
Group A:										
Keldseå (centre)	13							58	154	
Keldseå (margin)	14							50	147	
Gneiss, baked	Gn14b							51	148	
MEAN of A:	***A***	*3*	*28*	*42*	*2.994*	*6.9*	*322*	*53.2*	*149*	*7.1*
Group B:										
Dyke4	4							37	359	
Dyke5	5							-18	338	
Dyke6	6							-1	339	
Dyke9	9							-4	346	
Gneiss baked by Dyke9	Gn9b							-2	346	
Dyke11	11							25	327	
Dyke12, Salne	12							-21	321	
Vigehavn, Vi	Vi							15	159	
Vaseaa, Va	Va							-16	128	
Bølshavn, Bø(A)	Bø(A)							13	153	
MEAN of B:	***B***	*10*	*45*	*-41*	*9.173*	*15.3*	*11*	*-0.2*	*155*	*14.7*
Group C:										
Dyke1, Kås	1							46	62	
Dyke7, 7Th	7Th							53	59	
MEAN of C:	***C***	*2*	*82*	*75*	*1.999*	*9.1*	*749*	*50*	*60.6*	*15.9*
Other:										
Listed	Li							14	250	
Bølshavn, Bø(B)	Bø(B)							21	224	
Dyke2	2							-24	6	
Dyke3	3							64	178	
Dyke7	7							-15	27	
Dyke8	8							47	217	
Nexø Sandst.Fm. *)	NS *							55	203	11

*) Lower Cambrian Nexø Sandstone formation with angular standard deviation, obtained by great-circle analysis (Lewandowski & Abrahamsen, in preparation).

Mean pole B is the calculated mean of 10 poles: nos. 4, 5, 6, 9, 11, 12 and Gn9b (baked contact of dyke 9) as well as the poles of the Vaseå dyke (Va), the Vigehavn dyke (Vi) and the Bølshavn dyke (the A component). The poles cluster around the poles of the CSDG (central Scandinavian dolerite group), which covers an age interval between 1280 and 1150 Ma, but because of outliers like poles 4, 11, 12 (all with high α_{95} values) and Va, the scatter is higher, A95 = 14.7°. A baked contact test (positive) of the gneiss at dyke 9 confirms the primary origin of pole 9. Poles 11 and Va are situated some 30° SW of the CSDG main cluster. The data are less well constrained, but could suggest a somewhat prolonged igneous activity in the Bornholm area as compared to that of the CSDG (Fig. 7). As poles Bø (A), Vi and 5 are close to Silurian (northern) poles for Baltica, these poles may represent Silurian overprints as an alternative interpretation.

Mean pole C is the mean of only two poles: nos. 1 and 7Th (high-temperature component of dyke 7). The two poles are close to each other, but the value of A95 = 15.9° is rather high, due to the low number of poles. The poles are close to the Sveconorwegian Loop, suggesting an age around 950 Ma for dykes 1 and 7. Although Bornholm is situated only some 80 km south-east of the southernmost part of the Swedish Protogine Zone, the dykes on Bornholm do not appear to have been remagnetized by the Sveconorwegian overprint in contrast with the Swedish ones (Bylund 1992).

The ages of poles 2, 3, 7 and 8 remain enigmatic. They fall north of Equator in the area north of the Sveconorwegian Loop, and may suggest a northern prolongation of the late Precambrian APW curve via the preliminary pole NS of (?)Lower Cambrian age, to join the Lower Palaeozoic APW path of Fennoscandia, the reliability of the early Palaeozoic APW curve still being low (Torsvik *et al.* 1992). If this is correct, the presumed normal polarity of the Precambrian APW curve of Fennoscandia has to be reinterpreted as reverse, the poles in Fig. 7 then being south-seeking poles.

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Comparing the present data with other data from the Baltic Shield we find that in general there is a good correlation between the extensive data set defining the combined APW path of Baltica (Pesonen *et al.* 1989; Torsvik *et al.* 1992) and that of our more limited data set and the equivalent APW path of Bornholm. This indicates that Bornholm, although strongly faulted, has not been rotated as an individual microplate, but in a plate tectonic sense most probably has acted as a part of Baltica since the oldest dykes investigated were injected (i.e. at least since about 1340 Ma). This conclusion has important implications in relation to the interpretation of the remanent magnetization of the Lower Cambrian Nexø Sandstone Formation (Lewandowski & Abrahamsen, in preparation).

CONCLUSIONS

The present investigation, including principal component analysis of the characteristic remanent magnetization components found in 13 dykes of the N to NE trending dolerite dyke swarm(s) of Bornholm, has expanded and elaborated the results of an early investigation of four other dykes (Abrahamsen 1977).

By comparison between palaeomagnetic poles derived from the Bornholm dykes and the grand mean poles of the Precambrian APW path for Fennoscandia (Pesonen *et al.* 1989, 1991; Elming *et al.* 1993), three periods of Precambrian (intermediate to late Proterozoic) igneous/volcanic activities in Bornholm may be dated palaeomagnetically to the following groups:

A: around 1360 Ma (Subjotnian age),
B: around 1200 Ma (Jotnian age),
C: around 950 Ma (Sveconorwegian age).

Both geomagnetic polarities are present in the total group of 16 Precambrian dykes investigated, and the poles of A and B are both confirmed by positive contact tests to the host gneiss. The good fit onto the APW path of Fennoscandia suggests that individual rotations or major translations of Bornholm with respect of Fennoscandia or the Baltic Shield did not take place.

The poles not grouped in the clusters A, B or C, appear to fall on a smooth distribution, the course of which may indicate the shape of the (yet) poorly constrained prolongation of the Proterozoic APWP towards the Lower Palaeozoic APWP of Baltica. This part of the APWP is in fair accordance with the recently published path by Elming *et al.* (1993), except that the path suggested by us, from 850 Ma and onwards, is shifted further north (above 60° N) and crosses the Bering Strait, before joining the Lower Palaeozoic APWP (Fig. 7).

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